

Dear Hiring Manager,

I am writing to apply for the Graduate Engineer position advertised by Industrus Engineering. As a final-year master's student at UTS, I am eager to bring my technical skills and enthusiasm for climate-resilient infrastructure to your Graduate program. Please find my responses to the selection criteria below.

A commitment to ethical conduct and the highest standards of professional accountability

At Optik Consultancy, I have been assigned to forecast tidal inundation risk to council infrastructure for Tweed Shire Council under climate scenarios ranging from 2040 to 2120, and to deliver the resulting asset risk rankings to the client. While mapping risk-prone water and wastewater assets under sea-level rise, I identified that the Shared Socio-economic Pathway (SSP) data we had been provided to assess sea-level rise lacked accuracy around several areas of the council. Rather than proceed with the rankings as instructed, I raised the limitation with my team so that it could be properly accounted for in our assessment, even though this meant additional work to validate our approach against a tight delivery schedule. I believe that flagging a data limitation before it undermines a council's long-term planning decisions is a basic part of professional accountability, even when it is easier to work around a problem quietly.

Demonstrated ability to effectively communicate both with other engineers and with stakeholders from different fields

As part of my role at Optik Consultancy, I present weekly project updates to both my engineering supervisors and council stakeholders, who are themselves engineers. Because my internship is delivered as part of a UTS program funded by Cognizant, every Friday I am also exposed to a different set of industry professionals from outside engineering, presenting our project's progress to them in non-technical terms as practice for how we will eventually need to communicate with our client. During our final project handover stage, my team had to present our work to a non-technical supervisor from the business side of the organisation "Cognizant", who needed to understand not the engineering detail, but what our findings meant for the council's decision-making and budget planning. This regular shift between technical and non-technical audiences within each week has sharpened my ability to translate the same finding differently depending on who needs to act on it.

The ability to engage with a creative, innovative and proactive environment

Forecasting tidal inundation risk across a wide range of climate scenarios for the council meant working with data and modelling approaches that do not always fit a standard template. While sorting through the dataset, which contained over 30,000 assets, I recognised that reviewing each one individually would not be feasible within our timeline. Rather than working through the list linearly, I built a spatial filter in ArcGIS to group assets by exposure zone, allowing the team to assess risk at the zone level first and then drill down into individual assets only where the data indicated genuine concern. The rest of the team adopted this approach, which significantly reduced processing time while preserving the accuracy of the final rankings. The experience showed me that even as an intern, there is real room to shape how a project is approached rather than simply working through a task as instructed.

Demonstrated ability to use and manage information

Delivering asset risk rankings for the council involved working with climate projection data across multiple scenarios and time horizons and turning that volume of information into rankings that were both accurate and usable for council planning. As team lead, rather than deciding on a system unilaterally, I asked the team for input on how we could manage this information most efficiently given our deadline. We settled on tracking our

progress in a shared Excel workbook, organised by catchment area, so that each team member's findings could be cross-checked against the same structure rather than existing in separate, disconnected files. This gave us a single source of truth for the project, made it easy to spot gaps or inconsistencies as we went, and meant consolidating everyone's work into the final rankings took a fraction of the time it otherwise would have. Managing the information this way, rather than simply compiling it at the end, was what allowed us to deliver rankings the council could rely on.

The ability to manage own performance in a professional environment

Joining Optik Consultancy as my first engineering internship meant adjusting quickly to client deadlines and professional expectations, while continuing my studies alongside it. Early in the internship, it was quite a hassle to balance student life and work life, and as an international student, I also had to adjust to differences in workplace culture, including learning to use professional terminology instead of the more casual language I was used to from university group work. Rather than let this slow me down, I made a habit of noting how my supervisors phrased things in meetings and asked directly when I wasn't sure whether something was appropriate to raise or how to frame it. I also set clearer boundaries around my weekly schedule so that internship deadlines and coursework did not compete for the same hours. Over time, this made me noticeably more comfortable contributing in meetings and communicating with the team, and it taught me that managing my own performance often starts with being honest about what I don't yet know, rather than trying to appear confident before I actually am.

A demonstrated ability to work as part of a team and to show leadership when required

During the internship, I led a team of six final-year engineering students to deliver our project to the council. I delegated tasks based on individual strengths, tracked progress against the timeline, and was responsible for presenting our consolidated findings to supervisors and council stakeholders each week. One of the challenges I faced was holding the whole team accountable when someone fell behind on their expected contribution, particularly given that everyone was balancing the internship alongside their own coursework and commitments. Rather than letting this affect the wider team's morale or the project timeline, I checked in with the individual privately to understand what was causing the delay, adjusted the task split where needed, and set shorter interim check-ins so issues could be caught early rather than at the final deadline. This experience taught me that leadership in a junior role is less about giving direction and more about staying attentive enough to notice when someone needs support and addressing it before it becomes a bigger problem.

I would welcome the opportunity to discuss my application further at an interview. Please contact me at this email address Paromita.Chakma@student.uts.edu.au.

I look forward to hearing from you.

Kind Regards,

Paromita Chakma